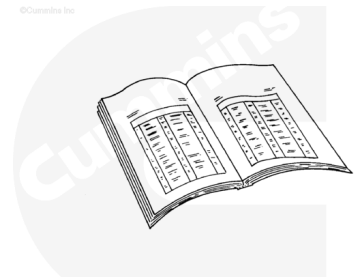


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Place a wooden block, the width of the oil pan, between the floor jack and oil pan to prevent damage to the engine.

- Remove the transmission, clutch, and all related components, if equipped. Refer to the OEM service manual.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/35/35-016-005-tr.html>)
- Use a floor jack or a suitable lifting fixture to support the rear of the engine.
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/35/35-013-020-tr.html>)
- Remove both rear engine mounts. Refer to Procedure 016-010 in Section 16.

(/qs3/pubsys2/xml/en/procedures/00/00-016-010.html)

- Remove the four oil pan rear mounting capscrews. Refer to Procedure 007-025 in Section 7.

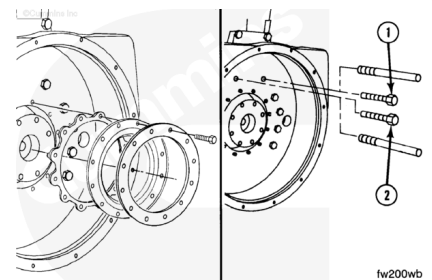
(/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)

- Remove the oil supply line from the flywheel housing.
- Remove the rear crankshaft oil seal. Refer to Procedure 001-024 in Section 1.

(/qs3/pubsys2/xml/en/procedures/101/101-001-024-tr.html)

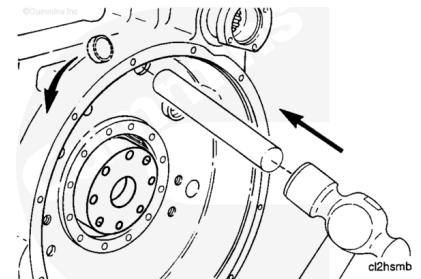
Remove

Remove capscrews (1 and 2) and install two guide pins, Part Number 3824929.



⚠ CAUTION ⚠

Do not attempt to back out or rotate the plugs from the housing. The cup plug bore will be damaged and oil leakage will occur.

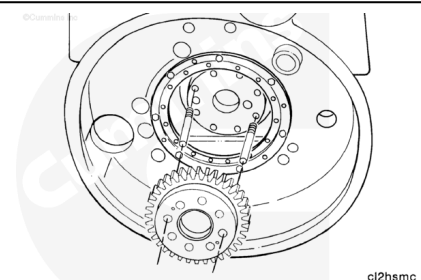


To gain access to the housing capscrews, use a cup plug driver, Part Number 3824927, to drive the cup plugs straight through into the housing.

Retrieve the plugs from inside the housing.

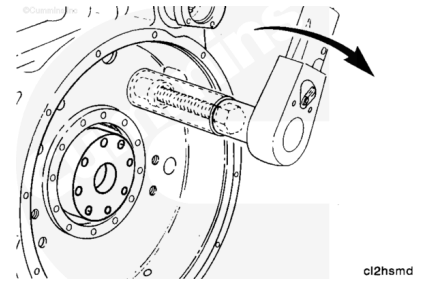
Remove the crankshaft drive gear.

Remove the two guide pins from the crankshaft.



⚠ CAUTION ⚠

When the idler shaft is removed, the idler gear can fall and become damaged. Hold the idler gear before pulling the idler shaft out.

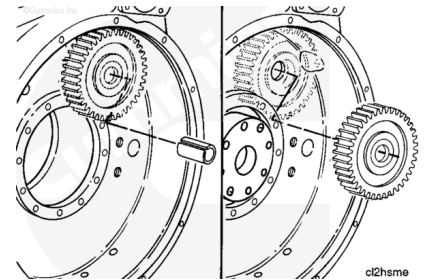


Remove the idler shaft retaining capscrew.

Use idler shaft puller, Part Number 3823709, to remove the idler shaft.

⚠ CAUTION ⚠

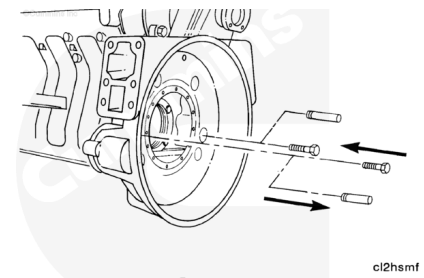
Do not drop the idler gear bearings as the idler gear is removed. Damage to the bearings can result.



Remove the idler gear to gain access to the remaining housing capscrews.

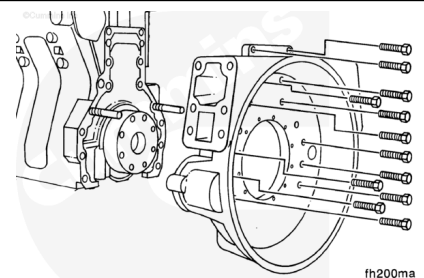
Remove two of the mounting capscrews, and install two guide pins, Part Number 3824930, to support the housing during removal.

Use offset wrench, Part Number 3824928, to remove the capscrews that are **not** in view. These capscrews are inside the housing.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the remaining capscrews.

Use a rubber hammer to loosen the flywheel housing.

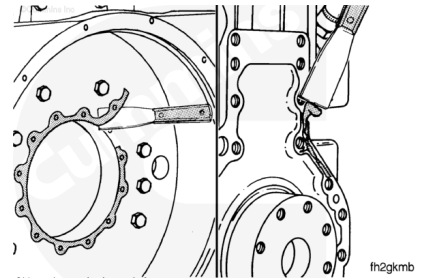
Note : Do not damage the rear of the oil pan gasket when removing the flywheel housing.

Remove the flywheel housing.

Clean and Inspect for Reuse

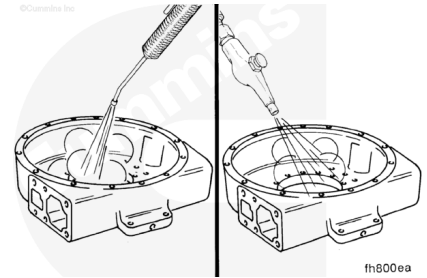
Remove the gasket material from the flywheel housing oil seal gasket surface.

Clean the sealant from the cylinder block surface.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

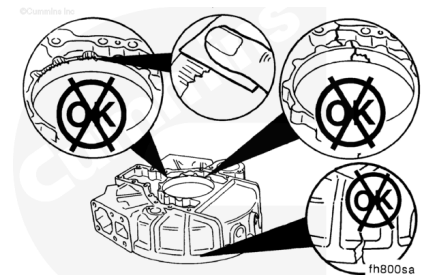
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the flywheel housing. Dry with compressed air.

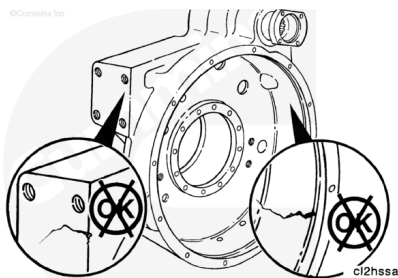
Inspect all surfaces for nicks, burrs, or cracks.

Use fine crocus cloth to remove small nicks and burrs.



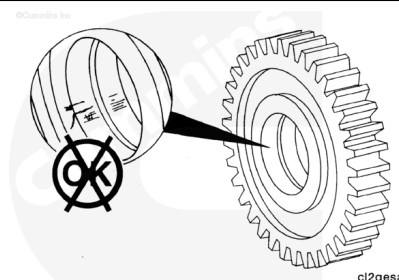
Inspect the rear engine power take off housing for cracks at the rear engine mounting surfaces and the flywheel bore.

Replace the housing if it is cracked.



Closely inspect the bore, side faces, and teeth of the rear engine power take off idler gear.

Replace the gear if there are cracks, discoloration from excessive heat, or other damage.

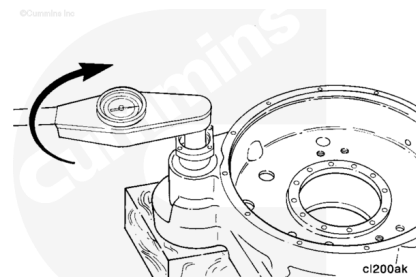


Check the output shaft rolling resistance with a torque wrench.

The rolling resistance **must** be between 0.6 to 1.1 N•m [5 to 10 in-lb].

If the resistance does **not** meet these specifications, add or subtract shims to obtain the correct rolling resistance.

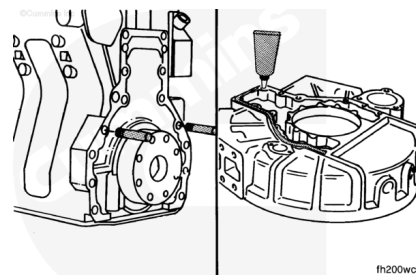
Adding shims will increase resistance; removing shims will decrease resistance. Any combination of shims can be used.



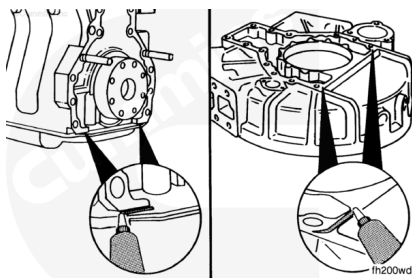
Install

Install two guide pins, Part Number 3376697, in the cylinder block to help support and align the housing during installation.

Apply a 2 mm [1/16 in] bead of silicone sealant, Part Number 3823494, on the mounting surface of the housing and around each capscrew hole.

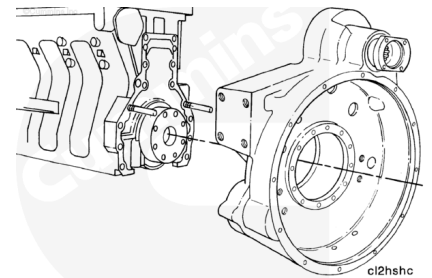


Apply silicone sealant to each rear corner of the oil pan gasket and flywheel housing mating surface.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a hoist when installing the rear engine power take off flywheel housing over the guide pins.

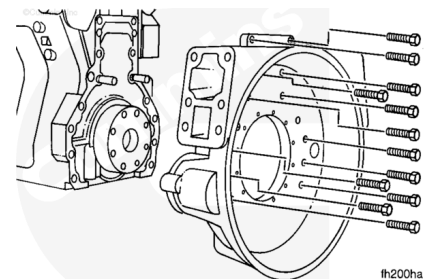
Install the flywheel housing over the guide pins.

Do **not** damage the oil pan gasket when installing the flywheel housing.

The flywheel housing bore **must** be aligned with the crankshaft. Do **not** tighten the capscrews to the final torque value until the flywheel housing is aligned.

Install the capscrews. Remove the guide pins, and install the remaining two capscrews.

Torque Value: 7 n·m [60 in-lb]



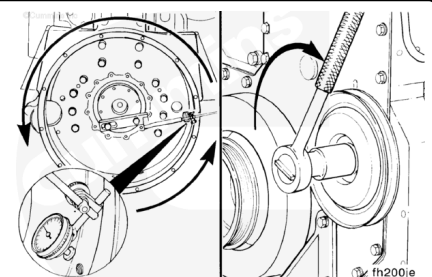
Measure

Bore Alignment

Use dial indicator gauge, Part Number 3376050, and dial gauge attachment, Part Number ST-1325, to measure the bore alignment.

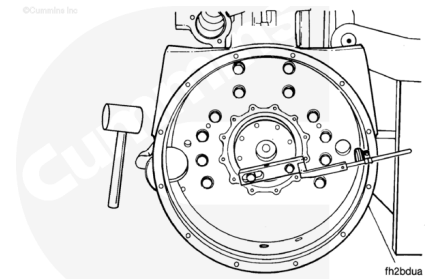
Adjust the dial indicator gauge to zero "0."

Use the accessory driveshaft to rotate the crankshaft one complete revolution (360 degrees).



Flywheel Housing Bore Alignment Maximum Total Indicator Reading

mm	SAE Number	in
0.30	00	0.012
0.25	0	0.010
0.25	1/2	0.010
0.20	1	0.008
0.20	2	0.008
0.20	3	0.008



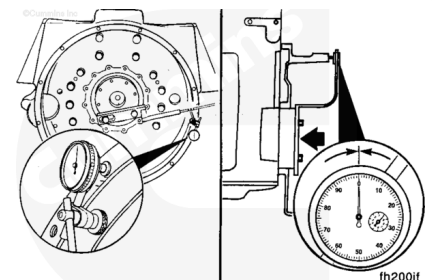
If the maximum bore alignment does **not** meet the specifications, use a rubber hammer to move the housing in the necessary direction, and repeat the measurement procedure.

If the bore alignment will **not** meet the specifications or the bore is **not** round, the housing **must** be replaced.

Face Alignment

⚠ CAUTION ⚠

The tip of the gauge must not enter the capscrew holes or the gauge will be damaged.



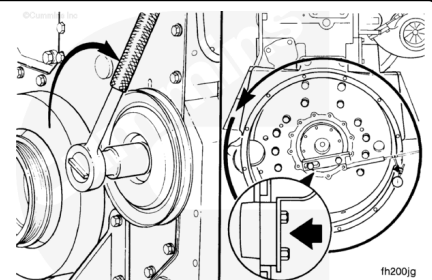
Install the contact tip of the dial indicator gauge against the flywheel housing face.

Push the crankshaft toward the front of the engine.

Adjust the dial indicator gauge to zero (0).

Monitor the dial indicator gauge while turning the crankshaft.

Use the accessory driveshaft to rotate the crankshaft one complete revolution (360 degrees) while keeping the crankshaft end thrust pushed forward.



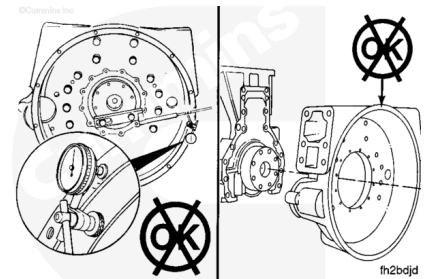
Flywheel Housing Face Alignment Maximum Total Indicator Reading

mm	SAE Number	in
0.30	00	0.012

Flywheel Housing Face Alignment Maximum Total Indicator Reading

mm	SAE Number	in
0.25	0	0.010
0.25	1/2	0.010
0.20	1	0.008
0.20	2	0.008
0.20	3	0.008

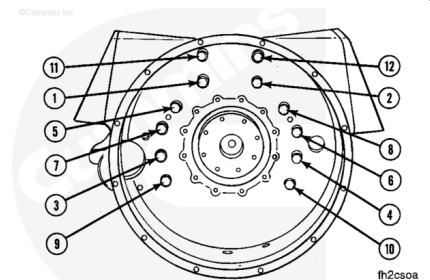
If the maximum face alignment does **not** meet the specifications, the flywheel housing **must** be replaced.



Tighten all capscrews in the sequence shown.

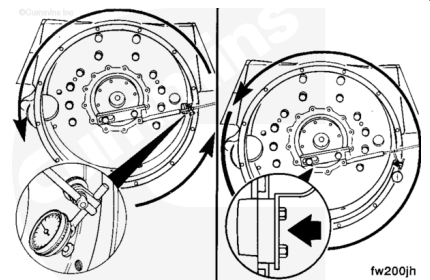
Torque Value:

1. 68 n•m [50 ft-lb]
2. 129 n•m [95 ft-lb]
3. 197 n•m [145 ft-lb]

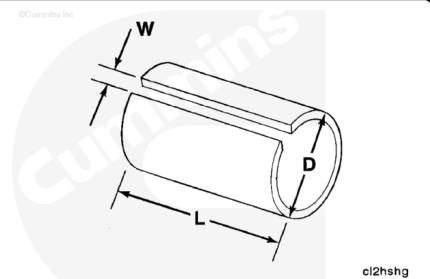


Measure the bore and face alignment again.

If bore and face alignment do **not** meet the specifications, loosen the housing capscrews. Tighten the capscrews, and measure the bore and face alignment again.



Manufacture a sleeve from 38.1-mm Outside Diameter [1.50-in] Outside Diameter (D) PVC, aluminum, thin wall hose, or equivalent, to the following dimensions:



Measurements

	mm	in
Length (L):	25.4	1.00

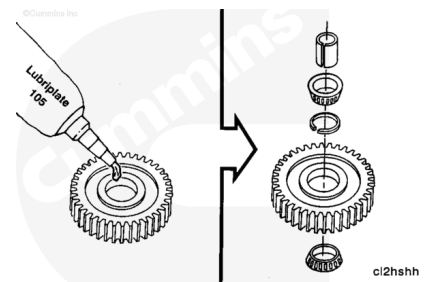
Measurements

	mm	in
Slot (W)	6.3	0.25

Apply Lubriplate™ 105, or equivalent, on the outer races and the bearings.

The outer bearing races of new replacement gears are already pressed into the gear.

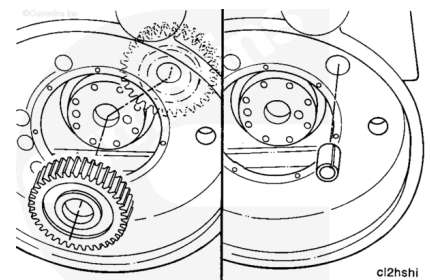
Install the bearing and spacer into the idler gear. Use the manufactured sleeve to hold the bearing assembly together when installing the idler gear assembly.



Apply a thin film of Lubriplate™ 105, or equivalent, into the idler shaft bore of the housing and on the idler shaft.

Install the idler gear assembly into the flywheel housing.

Hold the idler gear and bearings in place, and remove the plastic sleeve.

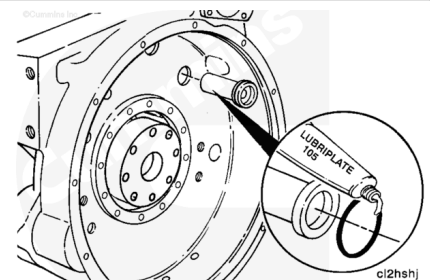


⚠ CAUTION ⚠

Do not use a hammer when installing the idler shaft and capscrew, or the part can be damaged.

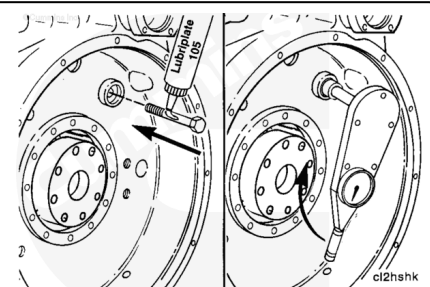
Use clean Lubriplate™ to lubricate the idler shaft o-ring, and install the o-ring onto the shaft.

Hold the gear assembly in place and insert the idler shaft through the housing and idler gear bearings.



Apply Lubriplate™ 105 under the head of the idler shaft capscrew. Insert the capscrew through the idler shaft. Tighten the installation capscrew with a torque wrench.

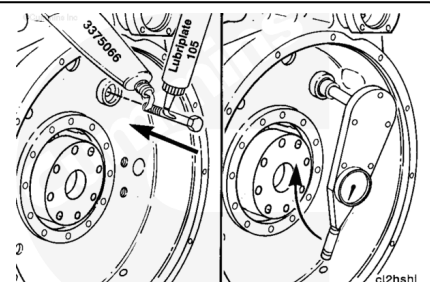
The torque needed to draw the idler shaft in place **must not** exceed 88 N•m [65 ft-lb]. If installation torque exceeds this amount, it indicates misalignment between the bore and the shaft. Remove the idler shaft and install it again.



After the idler shaft has been seated, remove the capscrew.

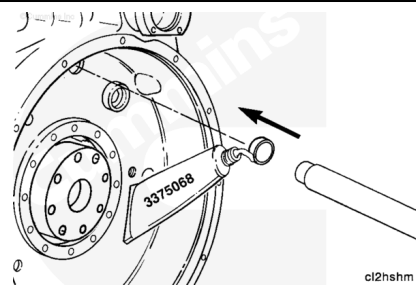
Apply pipe sealant, Part Number 3375066, to the threads of the idler shaft capscrew. Apply Lubriplate™ 105 under the head of the capscrew. Install the capscrew and tighten to its final torque value.

Torque Value: 102 n•m [75 ft-lb]

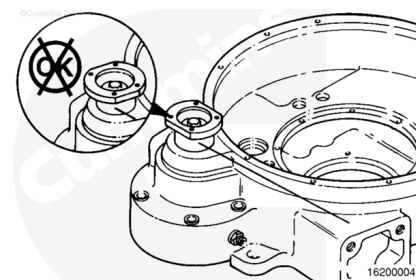


Apply a film of cup plug sealant, Part Number 3375068, to the outside diameter of the cup plug.

Use driver, Part Number 3823710, to install the cup plug into the housing, as shown.



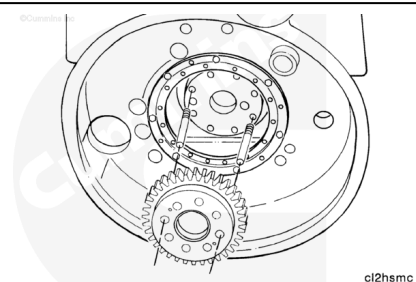
Turn the output flange so that the flat sides are on the top and bottom. This prevents any interference when the transmission is installed onto the housing.



Install two guide pins, Part Number 3824929, into the crankshaft flywheel mounting flange 180-degrees apart.

Make sure the crankshaft and crankshaft gear are clean.

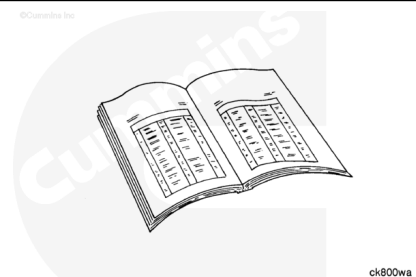
Install the crankshaft gear on the guide pins.



Finishing Steps

⚠ WARNING ⚠

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⚠ WARNING ⚠

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⚠ CAUTION ⚠

Place a wooden block, the width of the oil pan, between the floor jack and oil pan to prevent damage to the engine.

- Install the rear crankshaft oil seal. Refer to Procedure 001-024 in Section 1.
(/qs3/pubsys2/xml/en/procedures/101/101-001-024-tr.html)
- Install the four oil pan rear mounting capscrews. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)
- Install both rear engine mounts. Refer to Procedure 016-010 in Section 16.
(/qs3/pubsys2/xml/en/procedures/00/00-016-010.html)
- Install the starting motor. Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-020-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the floor jack or lifting fixture used to support the rear of the engine.
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/35/35-016-005-tr.html)
- Install the transmission, clutch, and all related components, if equipped. Refer to the OEM service manual.
- Operate the engine and check for proper operation.